GLOBAL HR FORUM
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Does AI Improve Productivity?

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Motivation

- Unlike previous technologies, GenAI is the first technology to generate new ideas and make decisions
- Thus, AI has the potential for significant productivity gains
- Also, the effects are likely to be heterogeneous across tasks and workers

Questions

- How prevalent is AI use?
- How will AI reshape work?
- How does AI affect workers’ productivity and time allocation at work?

What We Do

- AI Survey of Korean Workers

- Survey Design

  ✓ We construct a nationally representative survey of Korean workers
  ✓ Sample size: 5,512 respondents (aged 15-64)
  ✓ Survey period: May-June, 2025

- Post-stratification weighting applied based on:

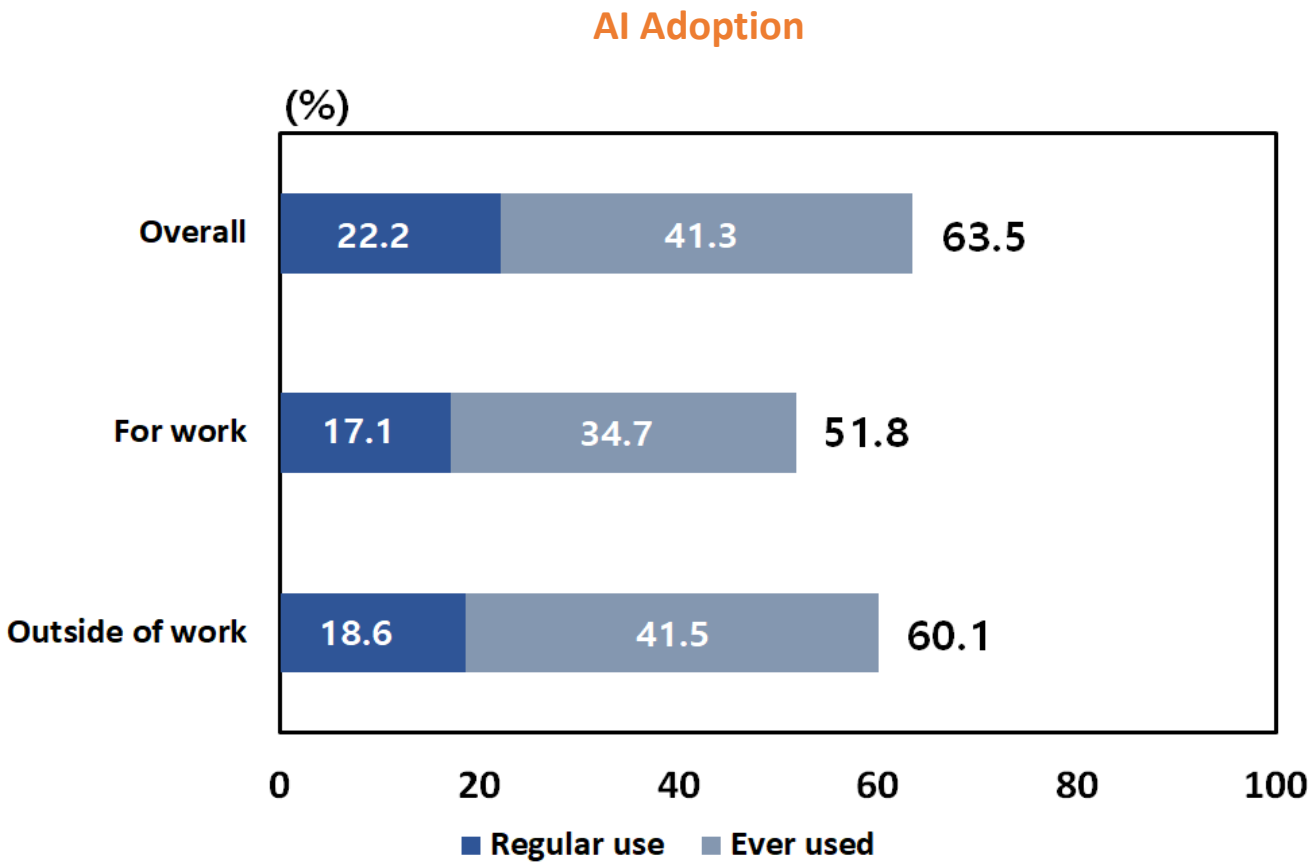
  ✓ Major occupational categories, age groups, gender
  ✓ Sample error: ±1.32%p at 95% confidence level

What We Find

- AI adoption is surging (51.8% for work)
- AI reduces working time by 3.8% (1.5 hours per week for a full-time worker)
- Time savings do not translate into higher output (time-output disconnect)
  - ✓ But productivity gains exist for some groups (self-employed, professionals)
- The future impact of AI depends less on technology and more on organizational design and incentives

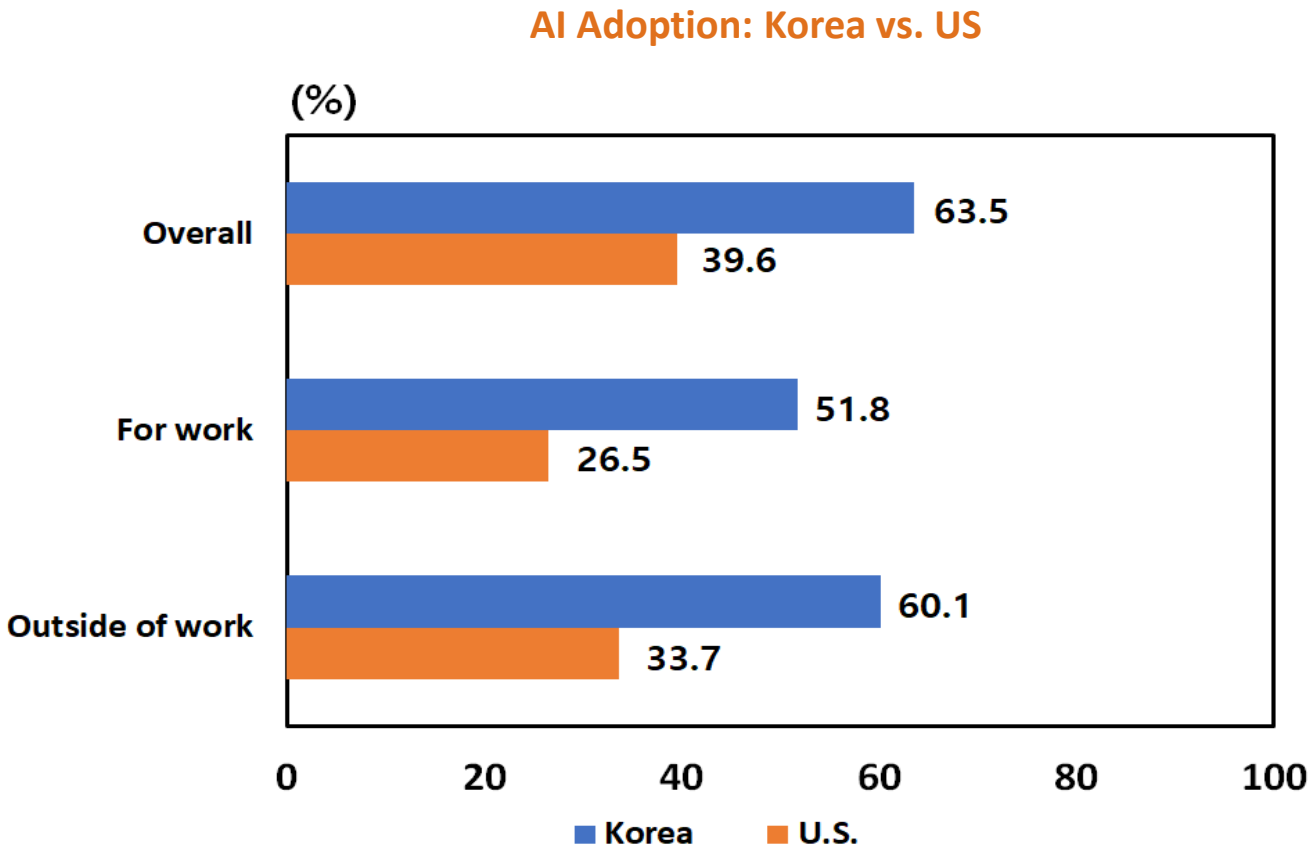
GenAI is Spreading Rapidly

More than half of Korean workers (51.8%) already use GenAI at work as of 2025



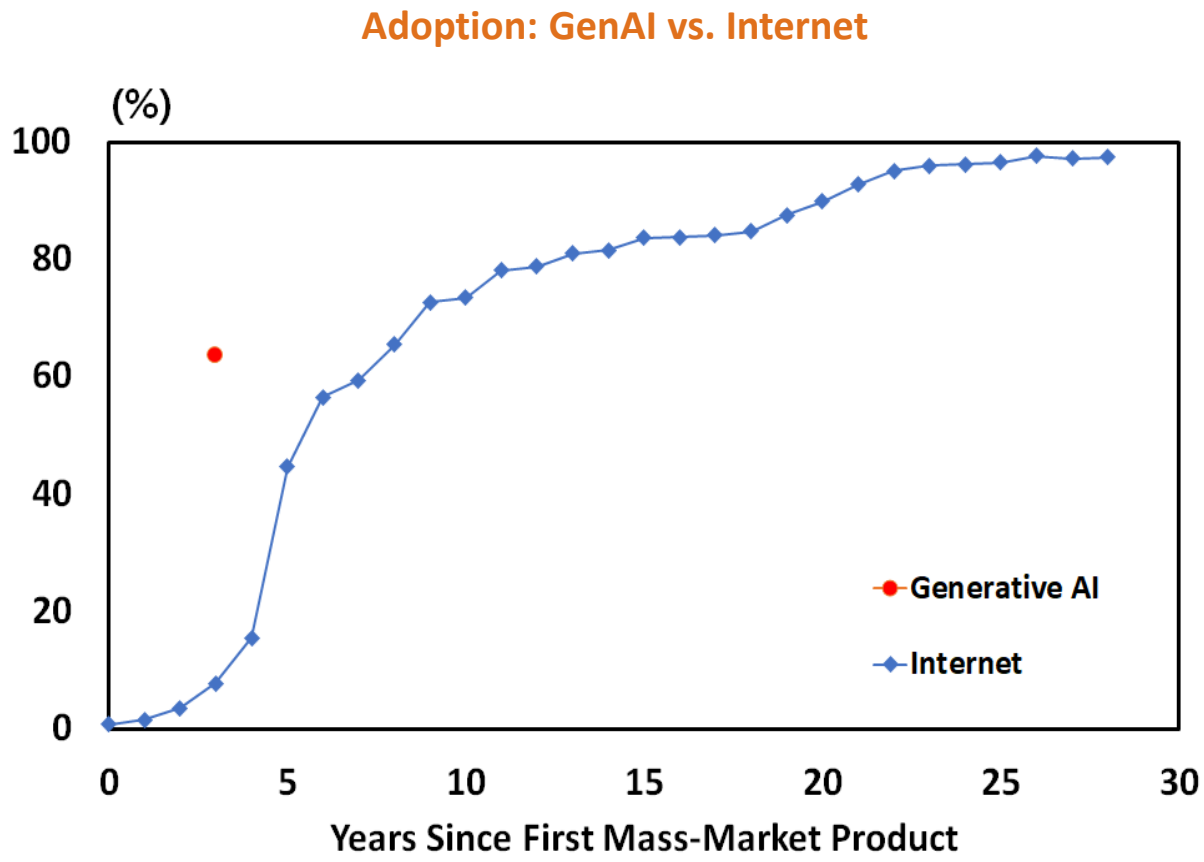
Korea vs. US

Korea shows 2x higher AI adoption rates than the US



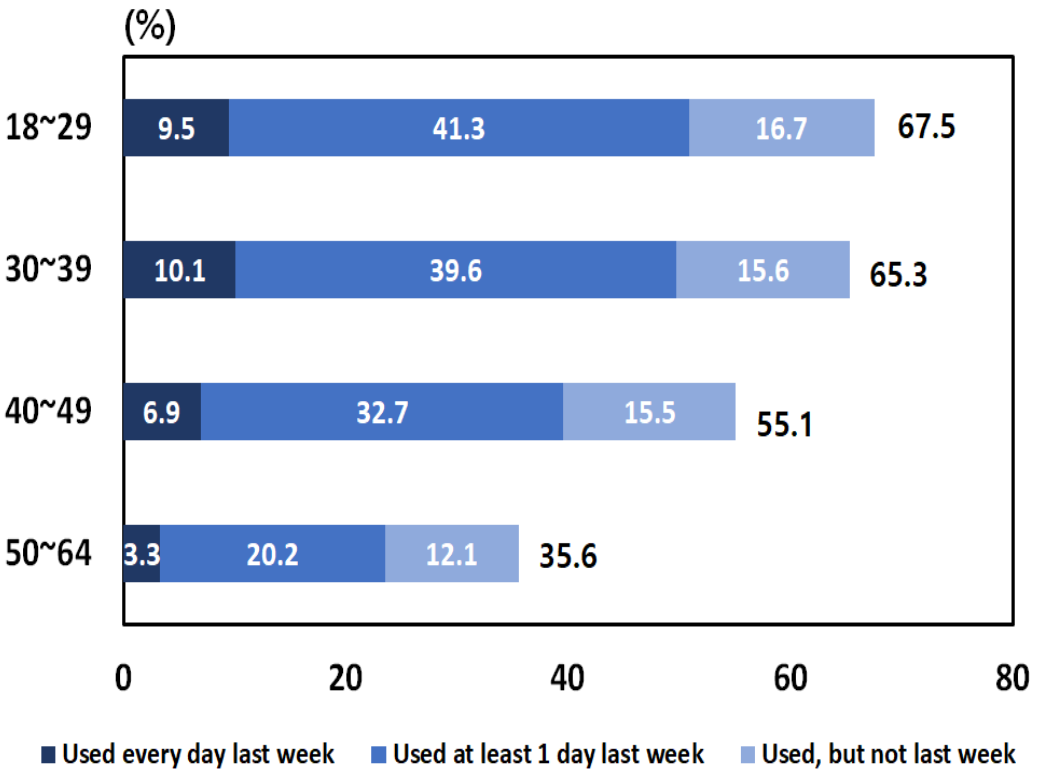
GenAI vs. Internet

GenAI adoption is 8 times faster than Internet adoption

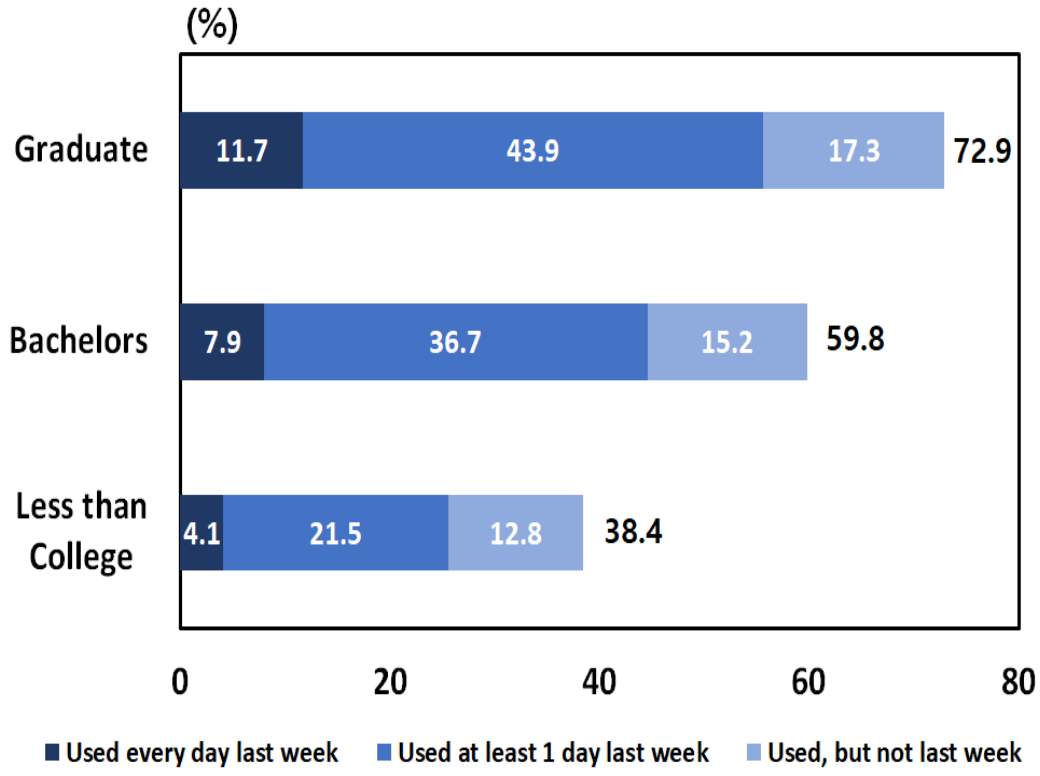


More Pronounced for Young & Highly Educated

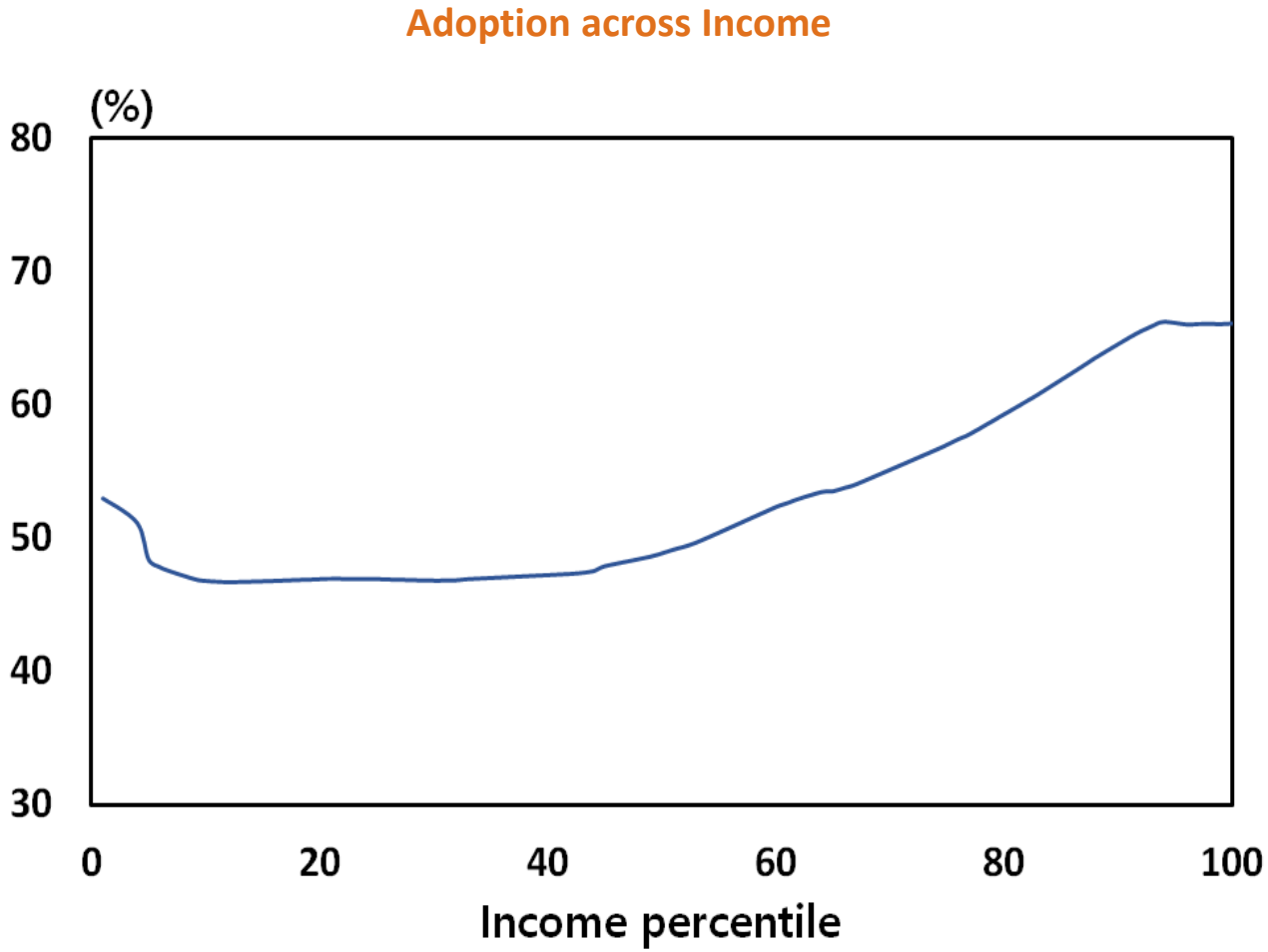
Adoption across Age Groups



Adoption by Education

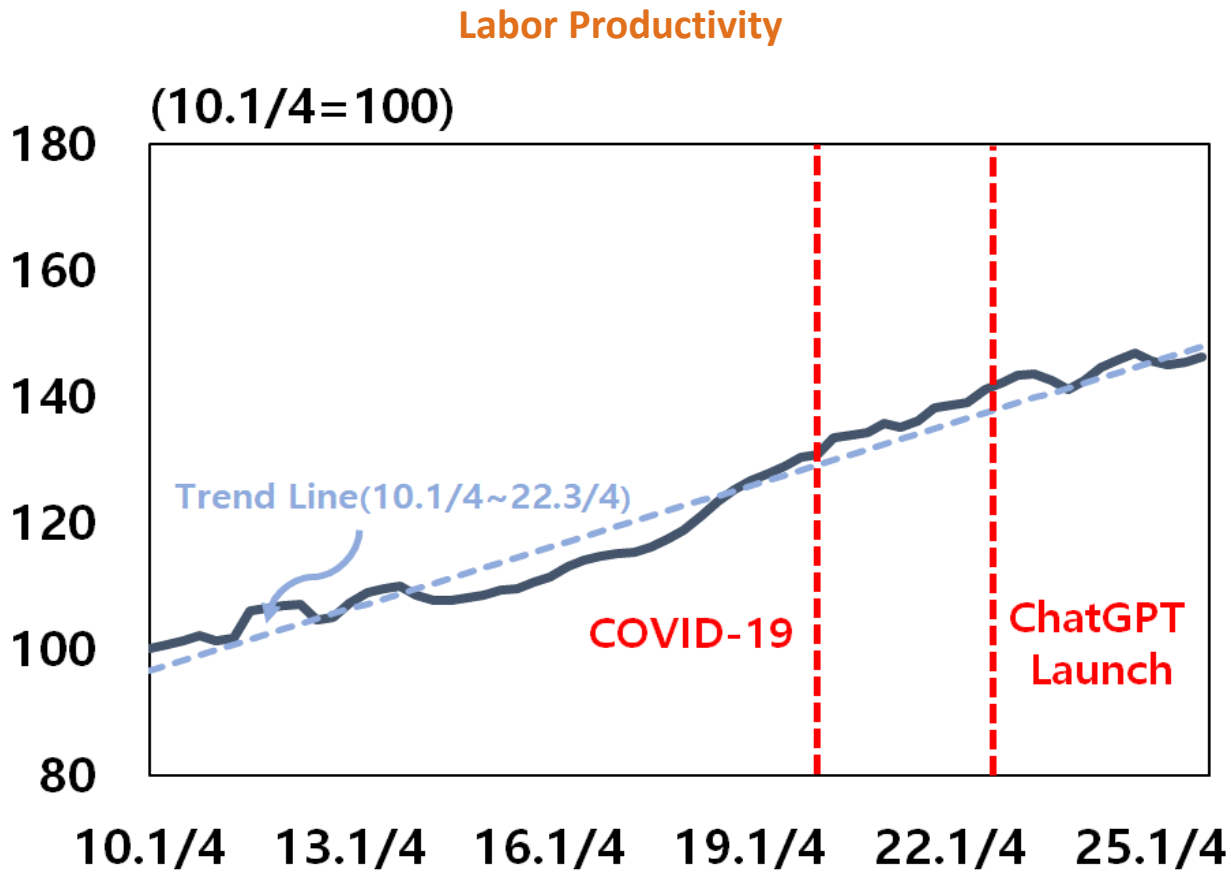


High-income Workers use GenAI More



AI Adoption is Surging, Productivity is Not

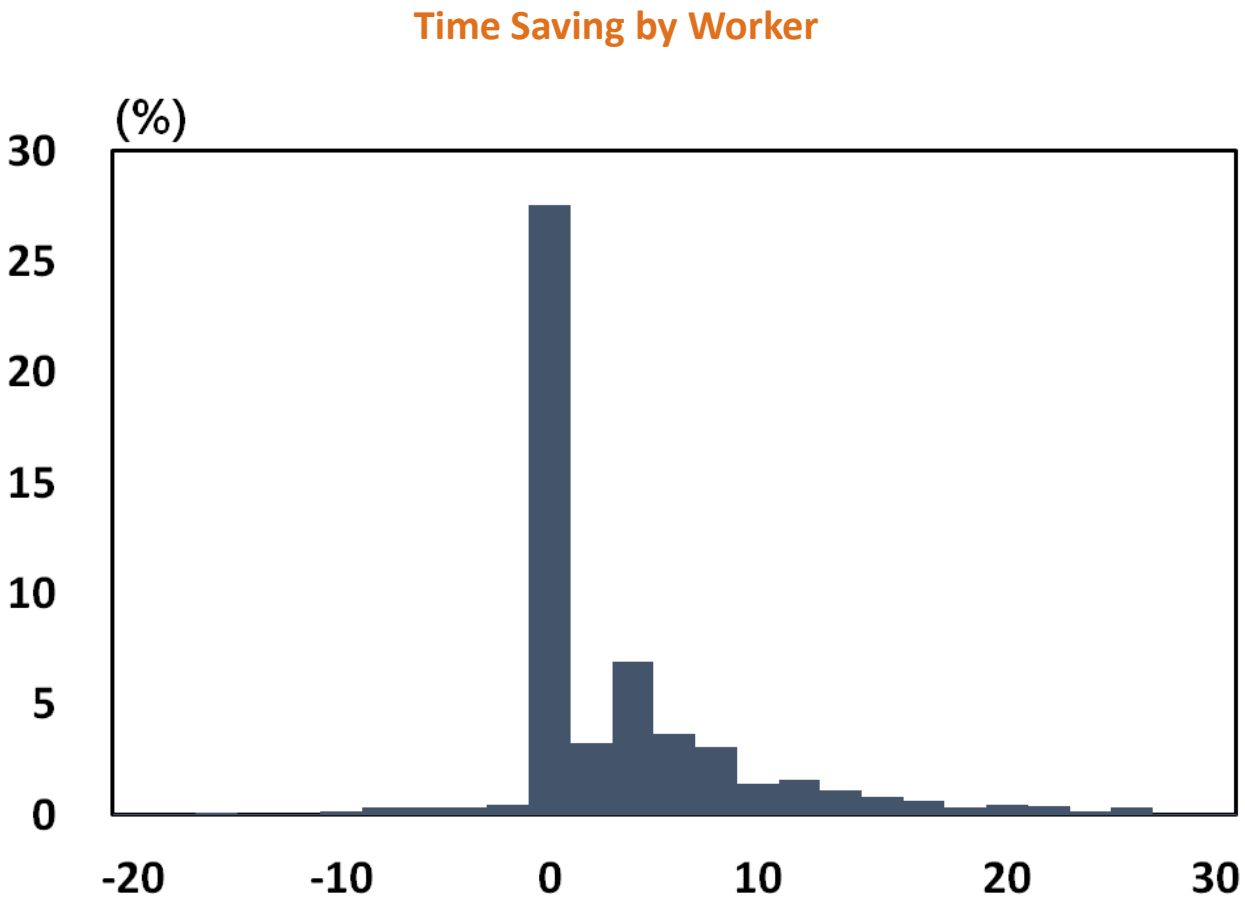
Labor productivity growth has shown little change over the past 3 years



Does AI Increase Productivity?

1. Does AI save working time?
2. If so, time savings lead to more production?

AI Clearly Saves Time



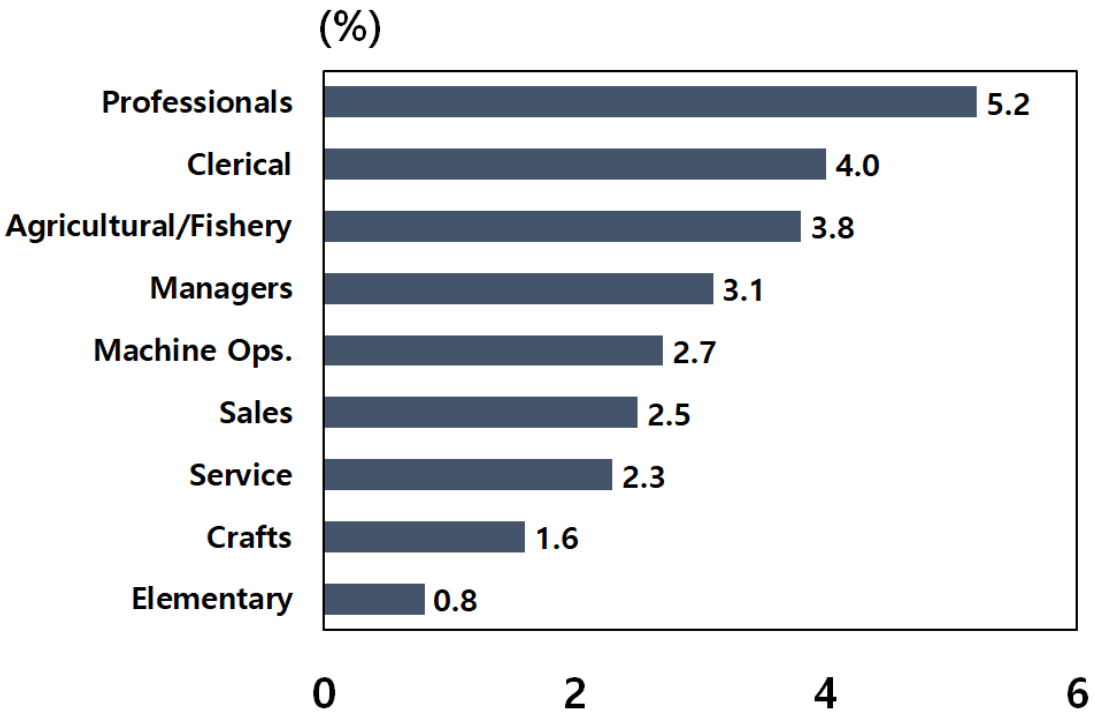
- Average working time falls by 3.8%
- Equivalent to about 1.5 hours per week for a full-time worker
- If all saved time were reallocated to productive work, potential productivity gains would be around 1.0% (upper bound)
- AI has already generated measurable efficiency gains

Time-saving Effects Differ

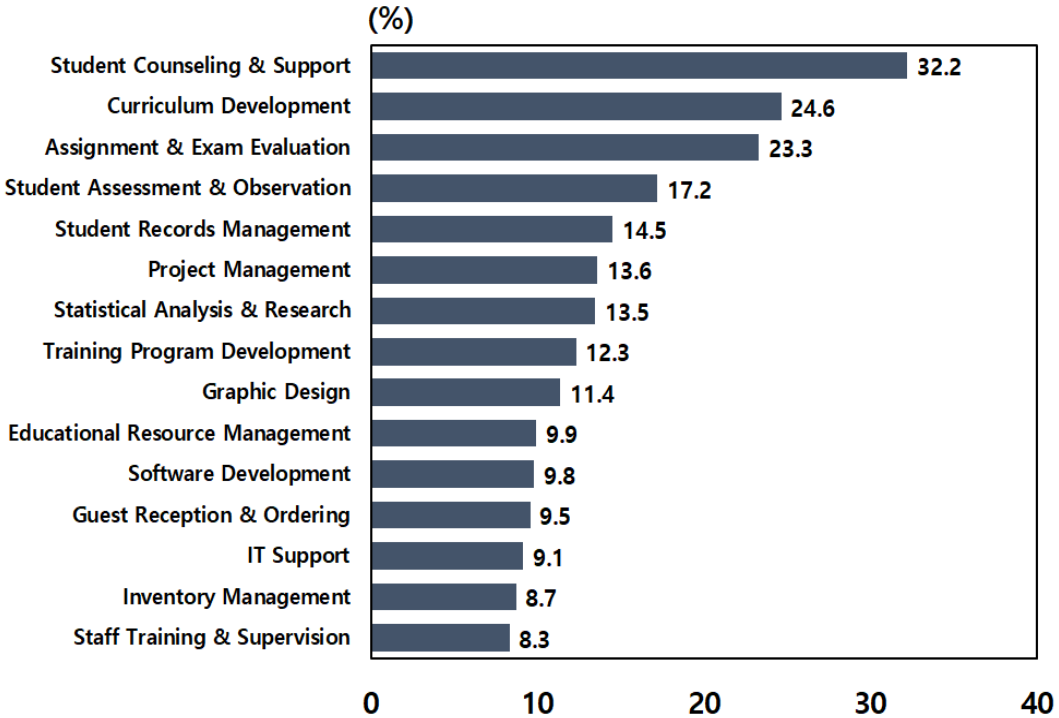
Stronger effects for professionals, clerks, managers

Larger gains for cognitive and non-routine tasks

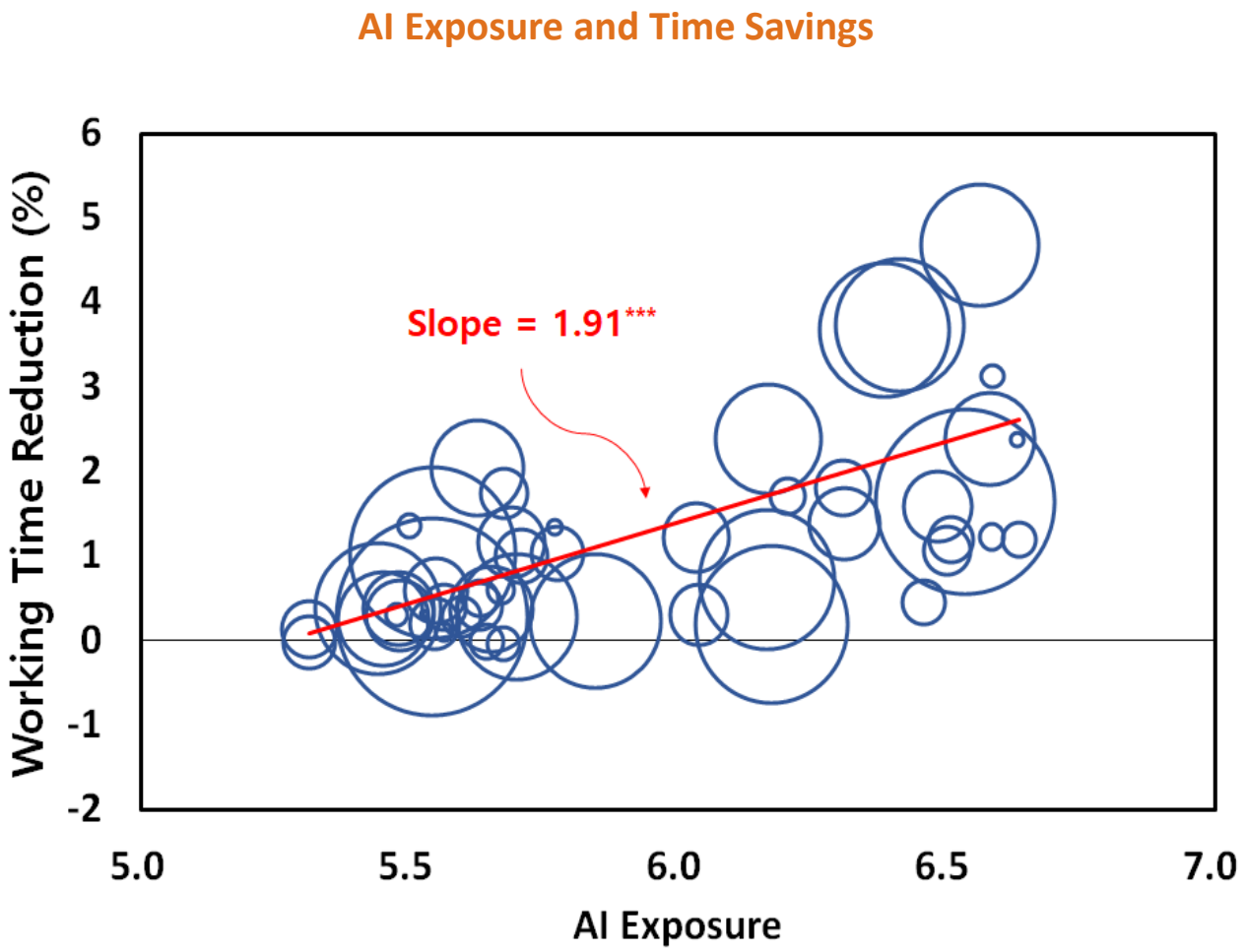
Time Saving by Occupation



Time Saving by Task

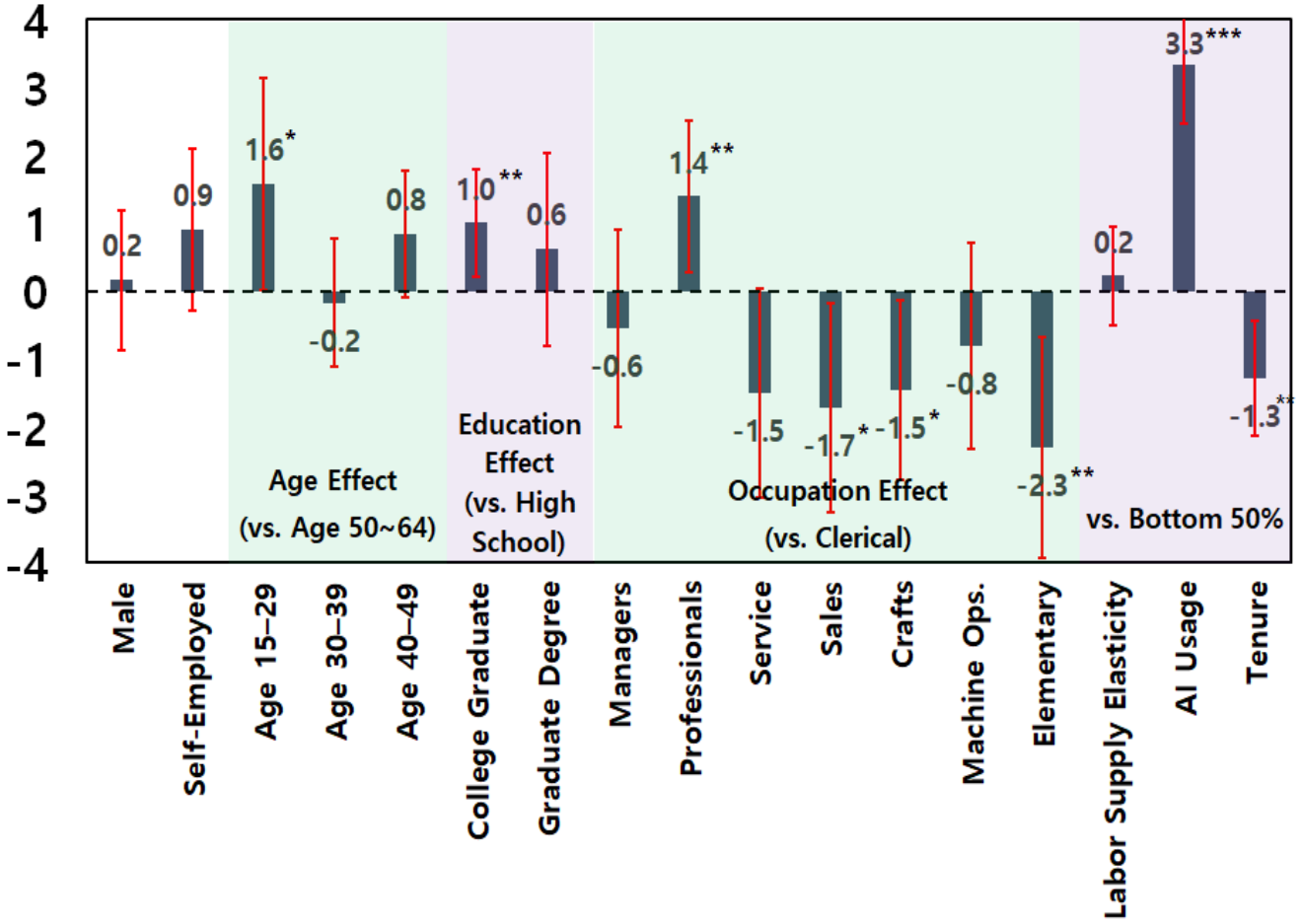


Highly Exposed Occupations Save More Time



Who Saves Time the Most?

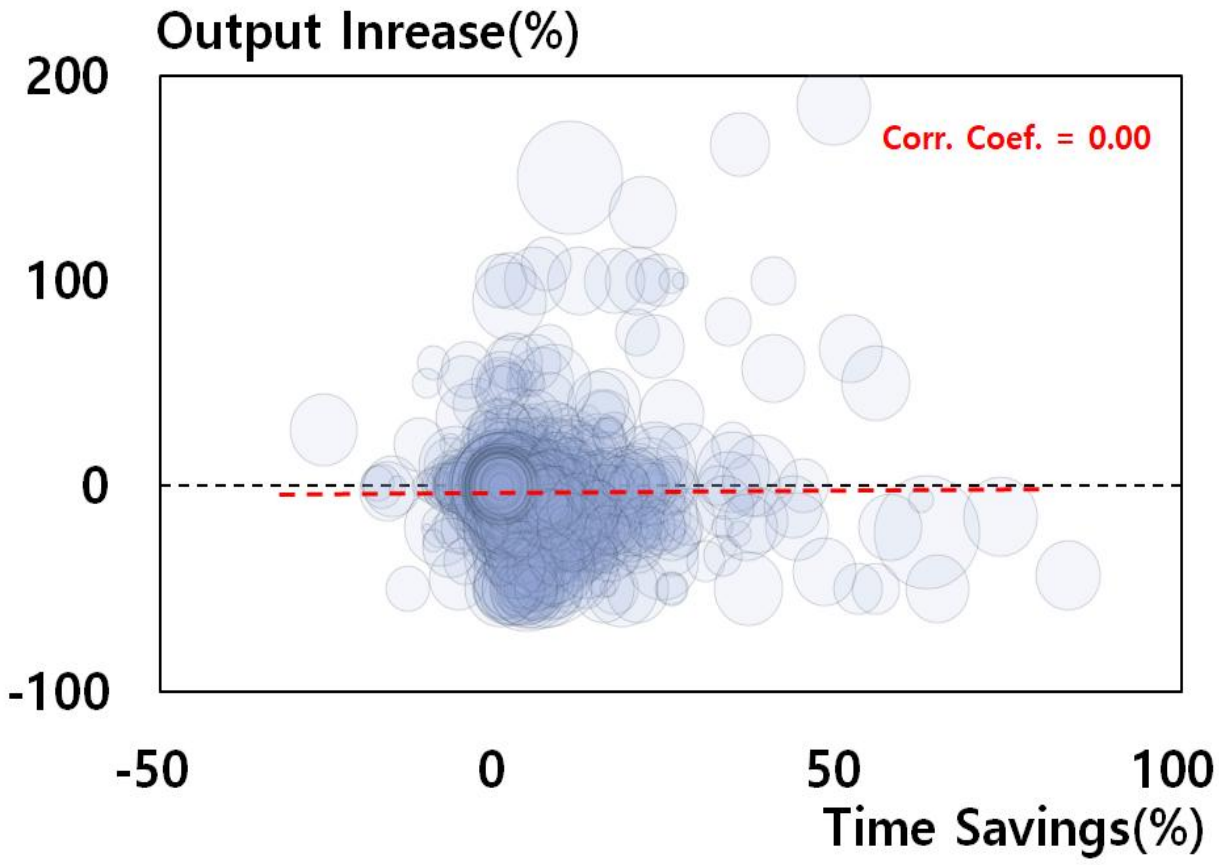
Regression Results



- Heavy AI users achieve larger time savings
- Less-experienced workers benefit more
- AI appears to reduce productivity gaps by compensating for experience deficits

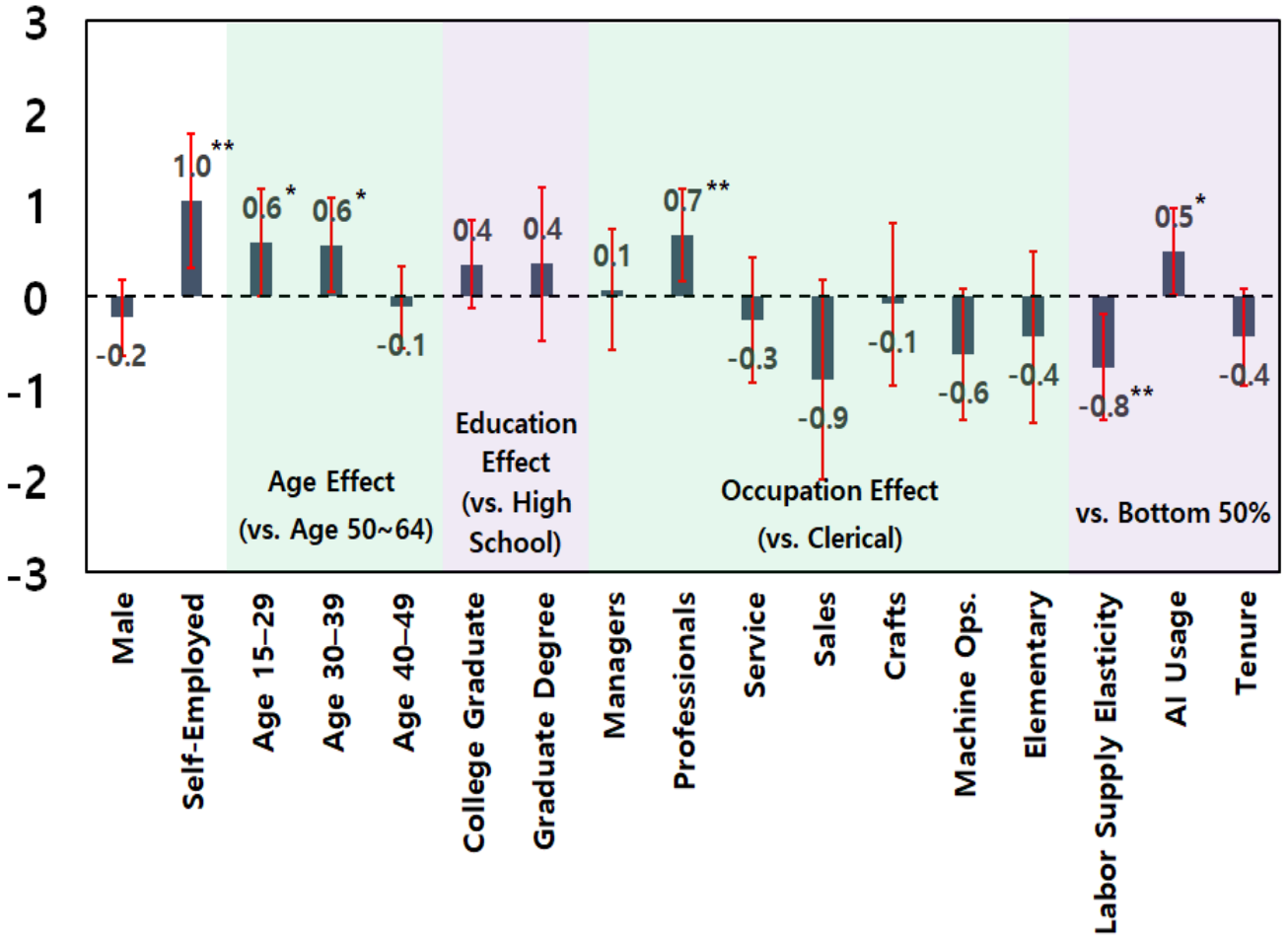
The Productivity Disconnect

- Workers save time, but firms do not yet produce more
- Virtually no relationship between time savings and output growth



But Productivity Gains Exist for Some Groups

Regression Results

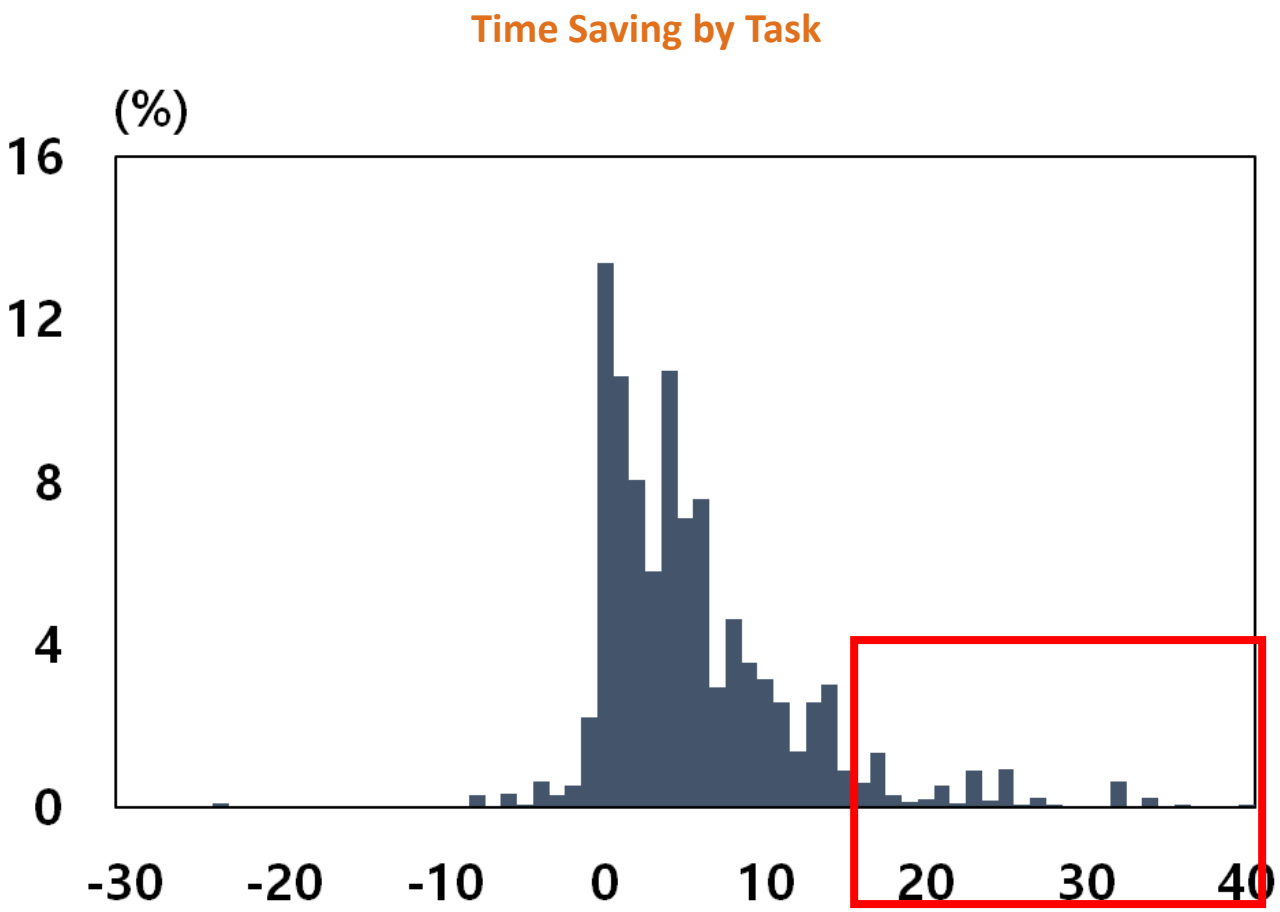


- Self-employed workers: stronger incentives to convert time savings into output
- Professionals: higher autonomy and visibility of performance
- Younger workers: greater adaptability to digital technologies
- Demographics matter less than incentives and task structure

Why Do Productivity Gains Take Time?

- Part of a natural J-curve (Solow paradox)
- The productivity disconnect may reflect a transitional phase commonly observed after the adoption of a general-purpose technology (e.g. electricity, computers, internet)
- History suggests that productivity gains often arrive with a lag
- The challenge is not diffusion itself, but translating efficiency gains into productivity gains

1. AI Adoption Remains at the Task Level



- Only a small share of tasks experience substantial (>20%) time savings
- AI improves selected tasks rather than entire jobs
- End-to-end automation remains limited

2. Organizations Have Not Yet Adapted

- AI is often layered onto existing processes
- Productivity depends on how tasks are connected, not just task efficiency (Demirer et al., 2026)
- Organizational culture and work processes matter as much as technology (Pereira et al., 2026)

3. Bottlenecks Still Matter

- Overall output is determined by the most constrained stage
- Faster analysis does not help if approval or coordination processes remain slow
- AI cannot eliminate productivity bottlenecks on its own

4. Misaligned Incentives

- Weak performance incentives reduce motivation to convert time savings into output
- AI may lower effort or distort behavior if incentives are poorly designed
- Productivity gains are strongest where performance and rewards are closely linked (e.g. among the self-employed and professionals)

Takeaways

- AI clearly saves time
- … but time savings have not yet translated into higher output
- The key challenge is no longer AI adoption, but converting efficiency gains into productivity gains

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EVERYTHING AI, 지속가능을 위한 길

EVERYTHING AI, The Path to Sustainability